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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025**25MCS114 / 25CYS111 – DESIGN THINKING**

(For MCS, CYS)

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Module – I & Module – II** each and **One full** question from **Module – III**.

Module – I

	* Marks	BT*	CO*	PO*
1. a) What is design thinking? Illustrate four of its pillars with examples.	10	2	1	1
b) Summarize the Design Thinking Skills required for a practitioner.	10	2	2	1
2. a) Explain five-step method for identifying customer needs by taking cordless screwdriver as an example.	10	2	1	2
b) Compare different types of product users with examples.	10	2	2	1
3. a) Explain the product development process.	10	2	1	2
b) Outline the need for People Centered Design with examples.	10	2	2	1

Module - II

4. a) Construct need metrics matrix with an example	10	3	3	1
b) Outline the process of establishing the target specifications with suspension fork example.	10	2	4	2
5. a) Summarize "Concept Development: The Front-End Process" with a neat diagram.	10	2	3	1
b) Interpret the interviewing as an art for eliciting customer needs data.	10	2	4	1
6. a) Explain how to Interpret Raw Data in Terms of Customer Needs with an example.	10	2	3	1
b) Interpret the steps for setting the Final Specifications of product.	10	2	4	2

Module - III

7. a) Outline the learning from Herman Miller story.	10	2	5	1
b) Identify and explain the environmental impacts in Product life cycle process.	10	2	5	3
8. a) Construct the flow chart for the Design for Environment Process (DFE) and explain.	10	2	5	1
b) Summarize the Internal and External Drivers of DFE process and the DFE objectives within the organization.	10	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
